Sumanthra Rao Yerrabelly

Software Engineer

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PROFESSIONAL EXPERIENCE

Graduate Research Volunteer, **University of Florida** **Sep 2025 - Present**

- Lead a research project on controllability and model editing in LLMs and vision-language models, building reproducible PyTorch and Hugging Face pipelines that enable systematic study of robustness, faithfulness, locality, and cross-modal representations, and provide the lab with a reusable framework for evaluating multimodal system behavior

Software Engineer, **Aiden AI** **Oct 2024 - Jul 2025**

- Built FastAPI and PostgreSQL backend services for a production AI product, conducting load tests across 100 concurrent connections to validate sub-150ms P95 latency, instrumented with Prometheus and Grafana for real-time request throughput monitoring.
- Developed and maintained RESTful APIs with FastAPI and PostGIS, implementing Redis caching and query optimization to reduce database load by 30% and enhance response times.
- Orchestrated asynchronous microservices using Redis, Celery, and Docker on AWS ECS, enabling real-time distributed task workflows with 99.98% uptime and horizontal scalability across multi-AZ deployments
- Integrated AutoGen agents and RAG pipelines with LangChain, QdrantDB, and Pydantic schema validation for claims decision support, automating inference tasks to save 120+ developer hours monthly and cut GPU inference costs by 25%.
- Partnered with ReactJS teams to build API-first integrations for real-time dashboards, increasing analytics adoption by 40%.
- Deployed Casbin RBAC access control and automated CI/CD pipelines leveraging GitHub Actions with staged rollouts and automated regression testing, achieving zero production rollbacks across 9 months of continuous feature delivery.
- Pioneered end-to-end data pipelines with structured logging and schema-versioned contracts, accelerating cross-team feature releases by 3x through standardized ETL workflows and automated integration testing.

Software Engineer Intern, **Aiden AI** **Jul 2024 - Sep 2024**

- Built enterprise-grade chatbots using LLaMA 3, ChromaDB, and LangChain with multi-tenant architecture and role-based access, delivering 95 percent accurate contextual responses.
- Designed a hybrid risk analysis engine with Graph Neural Networks (GNNs), Python, and REST APIs to process over 1M event logs, reducing unplanned downtime by 18 hours each quarter.
- Automated test workflows with GitHub Actions and pytest, enabling daily deployments with rapid patch turnaround and zero production rollbacks.
- Rebuilt the company website employing React, TailwindCSS, and Next.js, improving Lighthouse score to 96, reducing load time to 1.1 seconds, and accelerating release delivery by three weeks.

EDUCATION

University Of Florida - *Master's*, *Computer Science* **Aug 2025 - Aug 2027**

Mahindra University - *Bachelor's*, *Electronics And Computer Engineering* **Sep 2020 - Jul 2024**

PROJECTS

SUIT — Storage Unit Management System

- Overhauled the authentication layer to fix 7 OWASP security vulnerabilities, replacing plain-text password storage with bcrypt hashing, switching to JWT tokens stored in HttpOnly cookies, adding CSRF token validation, and enforcing input validation DTOs to block unauthorized role assignment on registration
- Built role-based access control using a three-stage middleware chain (AuthRequired, CSRFRequired, RoleRequired) in Go/Gin, restricting employee registration, role updates, and record deletion to manager accounts, with 6 access control tests covering all permission boundaries
- Implemented the core operations backend in Go/Gin with full CRUD for customers and units, a cross-entity search endpoint, automated daily lockout for overdue tenants after a 5-day grace period, a reservation system, and a move-out flow that clears unit assignments and cancels open reservations

AI-Guided Manufacturing Investigation Copilot

- Built a RAG-powered investigation copilot for semiconductor fab engineers using ChromaDB (bge-small-en-v1.5, 384-dim embeddings) with metadata-filtered similarity search, achieving sub-100ms retrieval latency across a 50-case knowledge base
- Designed a three-tier LLM fallback architecture (HuggingFace Inference API → Ollama local → deterministic rules) guaranteeing 100% response availability across connected, offline, and air-gapped deployment environments
- Implemented Pydantic v2 validation layer on all LLM outputs with case-insensitive normalization, type coercion, and automatic rule-based fallback on schema violation, eliminating unvalidated AI outputs from reaching end users
- Engineered a FastAPI REST backend with 4 endpoints, Power Apps custom connector integration (OpenAPI 3.1 → Swagger 2.0 conversion), and ngrok tunneling for enterprise deployment

SKILLS

Programming Languages : Python, JavaScript, SQL, Go, C#, Java

Frameworks & Libraries : FastAPI, Flask, ReactJS, Next.js, TailwindCSS, PyTorch, TensorFlow, LangChain, Hugging Face, LoRA, Scikit-learn, Spring

AI / ML & Data Science : Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, MLOps, Jupyter notebooks

Backend & API Development : RESTful APIs, Microservices Architecture, Async I/O, Celery, Redis, PostGIS, WCF, API Gateway, Authentication & Authorization (Casbin), Performance Optimization, object-oriented design, software architecture, scalable computing systems, secure software

Cloud & DevOps : AWS ECS, AWS EC2, Docker, GitHub Actions, GitLab CI/CD, Jenkins, Kubernetes, Prometheus, Grafana, ELK Stack, Monitoring & Observability

Databases : PostgreSQL, MongoDB, Redis, Neo4j, ChromaDB, Qdrant, SQL Server

Tools & Platforms : Weights & Biases, MLflow, DVC, Streamlit, Gradio, Casbin, Pytest

Collaboration & Methodologies : Agile/Scrum, Cross-Functional Teamwork, Continuous Integration/Deployment, Technical Documentation, Performance Benchmarking, Agile, SCRUM

ACHIEVEMENTS

Scholarships

Earned scholarships for academic years 2021–2022 and 2022–2023 at Mahindra University

Level 1 Certified Risk Manager

Achieved certification as a Level 1 Certified Risk Manager.